

DIGITAL WALLET – UPI PAYMENT SYSTEM

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Objective / Aim

To design and implement a Digital Wallet – UPI Payment System in Java using interfaces, classes, encapsulation, custom exceptions, String validation, and try-catch-finally exception handling.

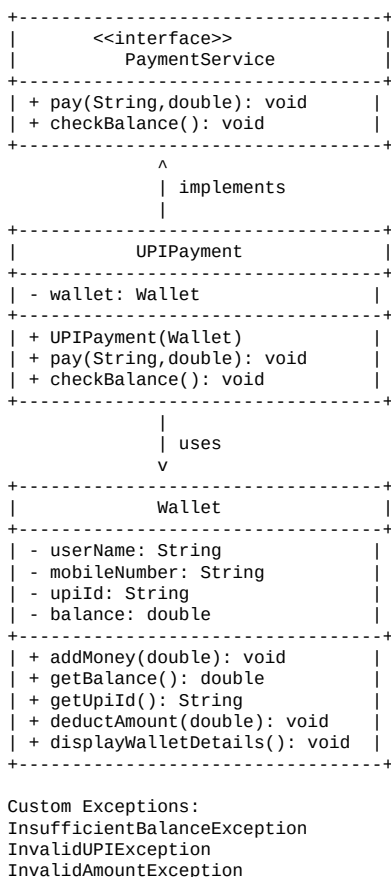
Algorithm / Logic

1. Create the PaymentService interface with pay() and checkBalance() methods.
2. Create custom exceptions for insufficient balance, invalid UPI ID, and invalid amount.
3. Create the Wallet class with private user details, UPI ID, and balance.
4. Define the Wallet constructor, addMoney(), getBalance(), getUpId(), deductAmount(), and displayWalletDetails().
5. Create the UPIPayment class implementing PaymentService.
6. Validate the receiver UPI ID using String operations.
7. Validate the payment amount and check the available wallet balance.
8. Process the payment and deduct the amount when all validations succeed.
9. Use try-catch-finally to handle transaction exceptions and display final wallet details.

Steps for Execution

1. Create the Java project and add PaymentService, exception classes, Wallet, UPIPayment, and DigitalWalletDemo.
2. Compile and run DigitalWalletDemo.java.
3. Create the Wallet object with the user details and initial balance.
4. Add Rs.2000 to the wallet.
5. Check the available wallet balance.
6. Make a UPI payment of Rs.1500 to friend@upi.
7. Observe the successful payment message and remaining balance.
8. Observe the final wallet details and final balance.

UML Class Diagram



Source Code

```
// Interface
interface PaymentService {
    void pay(String upiId, double amount)
        throws InvalidUPIException,
        InvalidAmountException,
        InsufficientBalanceException;
    void checkBalance();
}

// Custom Exception 1
```

```

class InsufficientBalanceException extends Exception {
    public InsufficientBalanceException(String message) {
        super(message);
    }
}

// Custom Exception 2
class InvalidUPIException extends Exception {
    public InvalidUPIException(String message) {
        super(message);
    }
}

// Custom Exception 3
class InvalidAmountException extends Exception {
    public InvalidAmountException(String message) {
        super(message);
    }
}

// Wallet Class
class Wallet {
    // Private data members - Encapsulation
    private String userName;
    private String mobileNumber;
    private String upiId;
    private double balance;

    // Constructor
    public Wallet(String userName, String mobileNumber,
        String upiId, double balance) {
        this.userName = userName;
        this.mobileNumber = mobileNumber;
        this.upiId = upiId;
        this.balance = balance;
    }

    // Add money
    public void addMoney(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Money added successfully: Rs. " + amount);
        } else {
            System.out.println("Invalid amount. Money cannot be added.");
        }
    }

    // Get balance
    public double getBalance() {
        return balance;
    }

    // Get UPI ID
    public String getUpiId() {
        return upiId;
    }

    // Deduct amount
    public void deductAmount(double amount) {
        balance -= amount;
    }

    // Display wallet details
    public void displayWalletDetails() {
        System.out.println("\n===== WALLET DETAILS =====");
        System.out.println("User Name : " + userName);
        System.out.println("Mobile Number: " + mobileNumber);
        System.out.println("UPI ID : " + upiId);
        System.out.println("Balance : Rs. " + balance);
    }
}

// UPI Payment Class
class UPIPayment implements PaymentService {
    private Wallet wallet;

    // Constructor
    public UPIPayment(Wallet wallet) {
        this.wallet = wallet;
    }

    // Payment method
    @Override
    public void pay(String upiId, double amount)
        throws InvalidUPIException,
        InvalidAmountException,
        InsufficientBalanceException {

        // Validate UPI ID using String operations
        if (upiId == null ||
            !upiId.contains("@") ||
            upiId.startsWith("@") ||
            upiId.endsWith("@")) {
            throw new InvalidUPIException(
                "Invalid UPI ID: " + upiId
            );
        }
    }
}

```

```

        );
    }

    // Validate amount
    if (amount <= 0) {
        throw new InvalidAmountException(
            "Invalid payment amount: Rs. " + amount
        );
    }

    // Check balance
    if (amount > wallet.getBalance()) {
        throw new InsufficientBalanceException(
            "Insufficient balance. Available balance: Rs. "
            + wallet.getBalance()
        );
    }

    // Deduct amount
    wallet.deductAmount(amount);
    System.out.println("\n===== PAYMENT SUCCESSFUL =====");
    System.out.println("Receiver UPI ID: " + upiId);
    System.out.println("Amount Paid : Rs. " + amount);
    System.out.println("Remaining Balance: Rs. "
        + wallet.getBalance());
    }

    // Check balance
    @Override
    public void checkBalance() {
        System.out.println(
            "Available Wallet Balance: Rs. "
            + wallet.getBalance()
        );
    }
}

// Main Class
public class DigitalWalletDemo {
    public static void main(String[] args) {

        // Create Wallet object
        Wallet wallet = new Wallet(
            "Rahul",
            "9876543210",
            "rahul@upi",
            5000
        );

        // Display initial wallet details
        wallet.displayWalletDetails();

        // Add money
        System.out.println("\n===== ADD MONEY =====");
        wallet.addMoney(2000);

        // Create UPIPayment object
        UPIPayment payment = new UPIPayment(wallet);

        // Check balance
        System.out.println("\n===== BALANCE CHECK =====");
        payment.checkBalance();

        // Payment transaction
        System.out.println("\n===== UPI PAYMENT =====");
        try {
            payment.pay("friend@upi", 1500);
        }
        catch (InvalidUPIException e) {
            System.out.println("Transaction Failed: "
                + e.getMessage());
        }
        catch (InvalidAmountException e) {
            System.out.println("Transaction Failed: "
                + e.getMessage());
        }
        catch (InsufficientBalanceException e) {
            System.out.println("Transaction Failed: "
                + e.getMessage());
        }
        finally {
            System.out.println("\nTransaction processing completed.");
        }

        // Display final details
        wallet.displayWalletDetails();

        // Final balance
        System.out.println("\nFinal Wallet Balance: Rs. "
            + wallet.getBalance());
    }
}

```

Output

```
===== WALLET DETAILS =====
User Name : Rahul
Mobile Number: 9876543210
UPI ID : rahul@upi
Balance : Rs. 5000.0

===== ADD MONEY =====
Money added successfully: Rs. 2000.0

===== BALANCE CHECK =====
Available Wallet Balance: Rs. 7000.0

===== UPI PAYMENT =====
===== PAYMENT SUCCESSFUL =====
Receiver UPI ID: friend@upi
Amount Paid : Rs. 1500.0
Remaining Balance: Rs. 5500.0

Transaction processing completed.

===== WALLET DETAILS =====
User Name : Rahul
Mobile Number: 9876543210
UPI ID : rahul@upi
Balance : Rs. 5500.0

Final Wallet Balance: Rs. 5500.0
```